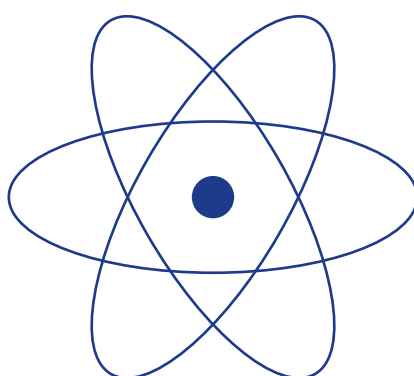


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Fluids

Olympiad Problem Set

Olympiad Training • Mechanics

SPhOH Team

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**. It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused collection of problems that develops fundamental problem-solving skills in **Fluid Mechanics and Statics**. The handout is **problem-centered**. Explanations are intentionally brief. The main purpose is to provide deliberate problem-solving practice.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

Difficulty Levels

The problems are divided into three training levels:

- **Level I** — Fundamental Olympiad problem-solving practice.
- **Level II** — Problems requiring combinations of ideas.
- **Level III** — More serious problems requiring deeper analysis.

Find more at: <https://devfhd.github.io/SPhOH/>.

Fluids

1.1 Units and Density Conversion

Topic Note: Units and Density Conversion

Density (ρ) is defined as mass per unit volume ($\rho = m/V$). In fluid mechanics, converting between SI units (kg/m^3) and CGS units (g/cm^3) is frequent:

$$1 \text{ g/cm}^3 = \frac{10^{-3} \text{ kg}}{(10^{-2} \text{ m})^3} = 1000 \text{ kg/m}^3$$

To convert from kg/m^3 to g/cm^3 , divide by 1000.

PROBLEM 1**LEVEL I****Density Unit Conversion I**

An object has a density of 200 kg/m^3 . What is its equivalent density expressed in g/cm^3 ?

PROBLEM 2**LEVEL II****Density Unit Conversion II**

If oil has a density of 600 kg/m^3 , let unit R be defined as $1 \text{ R} = \text{Oil Density}$. Express water's density in terms of R.

1.2 Hydrostatic Pressure and Pascal's Principle

Topic Note: Pressure and Pascal's Law

The pressure P exerted on a surface is force per unit area ($P = F/A$). In a static fluid of uniform density ρ , pressure varies with vertical depth h according to the hydrostatic equation:

$$P = P_0 + \rho gh \implies \Delta P = \rho g \Delta h$$

Pascal's Principle states that pressure applied to an enclosed incompressible fluid is transmitted undiminished throughout the fluid:

$$P = k = \frac{F_1}{A_1} = \frac{F_2}{A_2} \implies F_2 = F_1 \left(\frac{A_2}{A_1} \right)$$

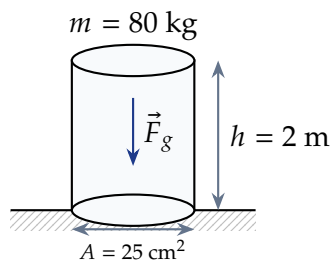
Where k is a constant throughout the fluid.

PROBLEM 3

LEVEL I

Pressure Under Heavy Metallic Cylinder

A uniform metallic cylinder with a mass of 80 kg and a height of 2 m rests vertically on one of its planar bases, which has a cross-sectional area of 25 cm^2 . What is the magnitude of the pressure exerted by the cylinder on the ground beneath it?

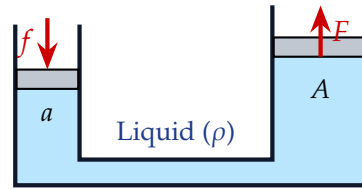


PROBLEM 4

LEVEL I

Force Scaling in Hydraulic Press

A piston of cross-sectional area a is used in a hydraulic press to exert a small force of magnitude f on the enclosed liquid. A connecting pipe leads to a larger piston of cross-sectional area A . What force magnitude F will the larger piston sustain without moving?

**PROBLEM 5**

LEVEL II

Circular Hydraulic Lift

A hydraulic lift consists of two circular pistons. The smaller piston has a diameter of 5.0 cm, while the larger piston has a diameter of 25.0 cm. How much force must be applied to the smaller piston to lift a 1500 kg vehicle? (Assume $g = 10 \text{ m/s}^2$)

PROBLEM 6

LEVEL II

Hydrostatic Pressure Difference and Depth

A vessel is filled with static fresh water ($\rho = 1000 \text{ kg/m}^3$). Two points within the water column, labeled Point A and Point B, are located such that Point B lies vertically lower than Point A. The pressure difference between these two points is measured to be 500 Pa. (Ignore atmospheric pressure, and assume $g = 10 \text{ m/s}^2$)

- (a) Determine which point (A or B) is at a higher hydrostatic pressure.
- (b) Calculate the vertical distance between Point A and Point B.
- (c) If Point A is located at a depth of $h_A = 0.10 \text{ m}$ below the water's surface, calculate the pressure at Point B.

PROBLEM 7

LEVEL II

Pressure Exerted by Chair Legs

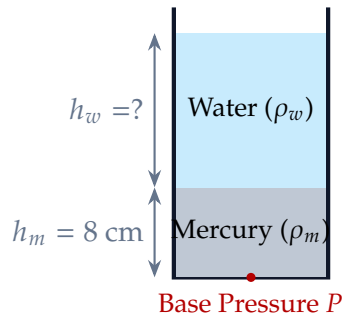
A large man sits on a four-legged chair with his feet off the floor. The combined mass of the man and chair is m . If the chair legs are circular and have a radius of r at the bottom, what pressure does each leg exert on the floor?

PROBLEM 8

LEVEL III

Adding Water over Mercury to Double Base Pressure

A tall cylinder with a cross-sectional area 12.0 cm^2 is partially filled with mercury ($\rho_m = 13.6 \text{ g/cm}^3$); the surface of the mercury is 8.00 cm above the bottom of the cylinder. Water ($\rho_w = 1.00 \text{ g/cm}^3$) is poured in on top of the mercury, and the two fluids don't mix. What volume of water must be added to double the pressure at the bottom of the cylinder? (Ignore atmospheric pressure)

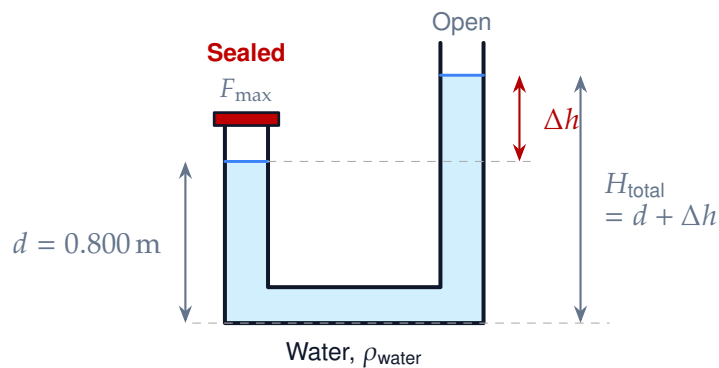


PROBLEM 9

LEVEL III

Critical Water Height in Sealed U-Tube

A plastic U-tube has a cross-sectional area of 5.00 cm^2 . The tube is filled with water until the short arm (of length $d = 0.800 \text{ m}$) is full. Then the short arm is sealed, and more water is gradually poured into the long arm. If the seal will pop off when the force on it exceeds 9.80 N , what total height of water in the long arm will put the seal on the verge of popping?



1.3 Buoyancy and Floating Bodies

Topic Note: Archimedes' Principle

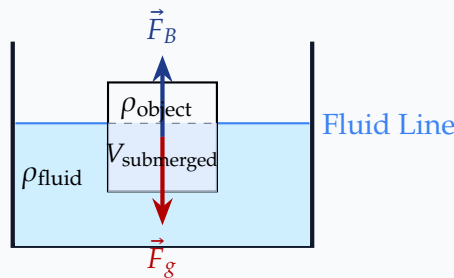
Archimedes' Principle states that a body wholly or partially submerged in a fluid is buoyed up by a force equal to the weight of the fluid displaced:

$$F_B = \rho_{\text{fluid}} V_{\text{submerged}} g$$

For a body floating in static equilibrium:

$$F_B = F_g \implies \rho_{\text{fluid}} V_{\text{submerged}} g = \rho_{\text{object}} V_{\text{total}} g \implies \frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

Notice that gravity g cancels out in floating equilibrium, making the submerged fraction independent of g .



PROBLEM 10

LEVEL I

Floating Object under Varied Gravity

A box is placed in a vessel filled with a liquid such that at equilibrium, one-fourth of the box is submerged in the liquid. If we take this whole system and place it on the surface of another planet that has twice the gravitational acceleration of Earth, what is the fraction of the submerged volume of the box?

PROBLEM 11

LEVEL I

Equilibrium Submerged Fraction

A solid object floats in static equilibrium in a container of water such that exactly half of its total volume is submerged beneath the surface. What is the ratio of the density of the object to

the density of water?

PROBLEM 12

LEVEL I

Emergent Height of Floating Cylinder

A uniform solid cylinder with a total height of 10 cm, a base area of 20 cm^2 , and a density of 0.3 g/cm^3 floats vertically in static equilibrium in a vessel filled with water ($\rho = 1.0 \text{ g/cm}^3$). What is the height of the portion of the cylinder extending above the water surface?

PROBLEM 13

LEVEL II

Submerged Ratio in Different Liquids

A uniform rectangular box floats upright first in water and then in oil. Express the ratio of its submerged height in water (h_w) to its submerged height in oil (h_o) in terms of the density of water (ρ_{water}) and the density of oil (ρ_{oil}).

PROBLEM 14

LEVEL II

Cavity Volume in Cast Metal

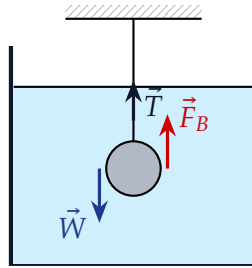
An iron casting containing a number of cavities weighs 6000 N in air and 4000 N in water. What is the total cavity volume in the casting? The density of solid iron is 7.87 g/cm^3 .

PROBLEM 15

LEVEL II

Tension in Submerged Metal Ball

A solid metallic ball of volume 1000 cm^3 is completely submerged in water and held in place by a rope attached to the ceiling. If the metal has a density of 2500 kg/m^3 , what is the tension in the rope? (Take $g = 10 \text{ m/s}^2$ and $\rho_{\text{water}} = 1000 \text{ kg/m}^3$)

**PROBLEM 16****LEVEL II****Density of Floating Hollow Cylinder**

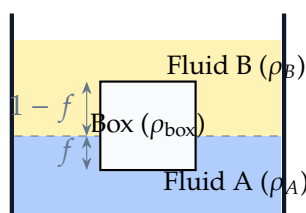
A hollow cylinder has an outer radius of 8.0 cm and an inner radius of 5.0 cm. It floats in a liquid with half of its volume submerged. If the density of the liquid is 0.65 g/cm^3 , what is the density of the material of the cylinder in g/cm^3 ?

PROBLEM 17**LEVEL II****Log Raft Capacity**

Three children, each of weight 356 N, make a log raft by lashing together logs of diameter 0.30 m and length 1.80 m. How many logs will be needed to keep them afloat in fresh water? Take the density of logs to be 800 kg/m^3 .

PROBLEM 18**LEVEL III****Floating Body at Two-Fluid Interface**

A solid box floats at the interface between two different fluids: fluid A (with density 1200 kg/m^3) and fluid B (with density 800 kg/m^3). If the density of the box is 900 kg/m^3 , what fraction of its volume is submerged in fluid A?



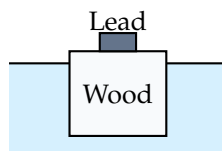
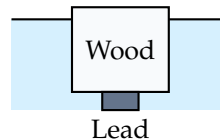
PROBLEM 19**LEVEL III****Dual Submersion Apparent Weight**

A rock is suspended from the lower end of a light string. When the rock is totally immersed in water ($\rho_w = 1.00 \text{ g/cm}^3$), the tension in the string is 1.20 N. When the rock is totally immersed in ethanol ($\rho_e = 0.79 \text{ g/cm}^3$), the tension in the string is 1.60 N. Calculate the weight of the rock.

PROBLEM 20**LEVEL III****Wood Block Weighted with Lead**

A wood block (mass 3.67 kg, density 600 kg/m^3) is fitted with lead (density $1.14 \times 10^4 \text{ kg/m}^3$) so that it floats in water with 0.900 of its volume submerged. Find the lead mass if the lead is fitted to the block's:

- (a) top (where lead is entirely in air).
- (b) bottom (where lead is completely submerged in water).

**(a) Lead on Top****(b) Lead on Bottom**

CHAPTER 2

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Density Unit Conversion I

Recall that $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. Divide the given density in kg/m^3 by 1000.

Hint 2 | Problem 2 – Density Unit Conversion II

Divide the density of water (1000 kg/m^3) by the oil density (600 kg/m^3).

Hint 3 | Problem 3 – Pressure Under Heavy Metallic Cylinder

Use $P = F/A = mg/A$. Convert the cross-sectional area from cm^2 to m^2 ($1 \text{ cm}^2 = 10^{-4} \text{ m}^2$).

Hint 4 | Problem 4 – Force Scaling in Hydraulic Press

Apply Pascal's principle: $\frac{f}{a} = \frac{F}{A}$. Solve directly for F .

Hint 5 | Problem 5 – Circular Hydraulic Lift

Express the ratio of piston areas in terms of their diameters squared: $A_2/A_1 = (d_2/d_1)^2$. Then apply Pascal's principle.

Hint 6 | Problem 6 – Hydrostatic Pressure Difference and Depth

Hydrostatic pressure increases linearly with depth ($\Delta P = \rho g \Delta h$). Point B is lower, so it experiences higher pressure.

Hint 7 | Problem 7 – Pressure Exerted by Chair Legs

Find the load on one leg ($F_{\text{leg}} = mg/4$) and divide by the contact area of a single circular leg (πr^2).

Hint 8 | Problem 8 – Adding Water over Mercury to Double Base Pressure

The initial bottom pressure is $P_{\text{initial}} = \rho_m g h_m$. Doubling the pressure requires the added water column to supply an additional hydrostatic pressure equal to P_{initial} , so $\rho_w g h_w = \rho_m g h_m$. Find

h_w and calculate volume $V_w = Ah_w$.

Hint 9 | Problem 9 – Critical Water Height in Sealed U-Tube

Find the maximum gauge pressure the seal can withstand ($P_{\max} = F_{\max}/A$). The gauge pressure at the seal is equal to the pressure exerted by the difference in water levels between the two arms, $\rho g \Delta h$. The total height in the long arm will be $d + \Delta h$.

Hint 10 | Problem 10 – Floating Object under Varied Gravity

Set up the vertical equilibrium equation $F_B = mg$. Observe that g cancels out on both sides.

Hint 11 | Problem 11 – Equilibrium Submerged Fraction

Equate buoyant force and total weight: $\rho_{\text{obj}} V_{\text{tot}} g = \rho_w V_{\text{sub}} g$. Solve for ρ_{obj}/ρ_w .

Hint 12 | Problem 12 – Emergent Height of Floating Cylinder

First calculate submerged height $h_{\text{sub}} = (\rho_{\text{cyl}}/\rho_w)h_{\text{tot}}$, then compute emergent height $h_{\text{above}} = h_{\text{tot}} - h_{\text{sub}}$.

Hint 13 | Problem 13 – Submerged Ratio in Different Liquids

Since the total weight of the box is constant, the buoyant forces in water and oil must be equal: $\rho_w h_w A g = \rho_o h_o A g$.

Hint 14 | Problem 14 – Cavity Volume in Cast Metal

The buoyant force $F_B = W_{\text{air}} - W_{\text{water}} = 2000 \text{ N}$ gives total external volume $V_{\text{ext}} = F_B/(\rho_w g)$. The actual volume of iron is $V_{\text{iron}} = W_{\text{air}}/(\rho_{\text{iron}} g)$. Total cavity volume is $V_{\text{ext}} - V_{\text{iron}}$.

Hint 15 | Problem 15 – Tension in Submerged Metal Ball

Draw a free-body diagram: upward tension T plus buoyant force F_B balance downward weight W , giving $T = W - F_B = (\rho_m - \rho_w)Vg$.

Hint 16 | Problem 16 – Density of Floating Hollow Cylinder

Equate buoyant force $\frac{1}{2}\pi R_{\text{outer}}^2 L \rho_{\text{liquid}} g$ to weight $\pi(R_{\text{outer}}^2 - R_{\text{inner}}^2)L \rho_{\text{mat}} g$.

Hint 17 | Problem 17 – Log Raft Capacity

Calculate the volume of a single cylindrical log $V = \pi r^2 L$. For the raft to stay afloat when fully submerged, the maximum buoyant force on N logs must equal or exceed the combined weight of the logs and the children: $N \rho_w V g \geq N \rho_{\text{log}} V g + 3W_{\text{child}}$. Solve for the minimum integer N .

Hint 18 | Problem 18 – Floating Body at Two-Fluid Interface

Let f be the fraction in fluid A, so $(1 - f)$ is in fluid B. Write the total buoyant force $fV\rho_A g + (1 - f)V\rho_B g = V\rho_{\text{box}}g$ and solve for f .

Hint 19 | Problem 19 – Dual Submersion Apparent Weight

Write tension equations $T_w = W - \rho_w Vg$ and $T_e = W - \rho_e Vg$. Subtract them to solve for Vg , then substitute back to find W .

Hint 20 | Problem 20 – Wood Block Weighted with Lead

For part (a), only the submerged volume of the wood provides buoyancy: $\rho_w(0.900V_{\text{wood}})g = (m_{\text{wood}} + m_{\text{lead}})g$. For part (b), both the 0.900 volume of wood and the full volume of lead contribute to buoyancy: $\rho_w(0.900V_{\text{wood}} + V_{\text{lead}})g = (m_{\text{wood}} + m_{\text{lead}})g$.